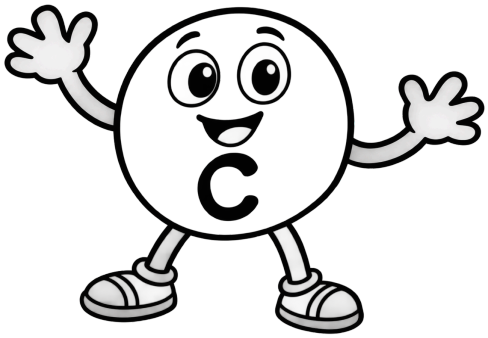
Name: _____ Period: _____

Earth and Space Science

Trace Carl the Carbon Atom

The Challenge

Millions of years ago, Carl the Carbon Atom was part of a tree growing in the area that is now Great Smoky Mountains National Park. Today, Carl is stuck in the limestone of Carlsbad Caverns National Park, which formed from an ancient reef about 265 million years ago. How did Carl get there?



Directions

1. With your partner, lay out the cards to build Carl's path. Start with the tree and end with the limestone. Put a **process** card between each pair of **location** cards. Keep the reading closed and use what you remember.
2. Record your path in the table. Add the **sphere** for each stop.
3. Open yesterday's reading (paragraphs 1 and 2). Check your path and fix anything in a different color.

Stop	Where is Carl?	Sphere	Process that moves Carl to the next stop
1	In the wood of a tree	Biosphere	
2			
3			
4			
End	In the limestone of Carlsbad Caverns		(end of path)

1. Which process cards did Carl **not** need? Choose one and explain why it does not help Carl get from the tree to the limestone.

2. Which stops are part of the **slow** carbon cycle? About how long could Carl be stuck in the limestone?

3. Which process card could finally get Carl out of the limestone and back into the atmosphere?

4. Write one sentence in Carl's voice about the slowest part of his trip.